

FRONTIER CLASS

Running GLM-5.2: locally and in the cloud

Only about 40B parameters are active per token, so the weights have to be held, not computed. The FP8 checkpoint is 744 GB: eight 96 GB cards in a 4U server locally, eight B200s in the cloud. A two-card workstation runs a 4-bit copy.

Model: GLM-5.2 · 744B MoE, ~40B active per token · MIT · 1M tokens · FP8 ≈ 744 GB · 4-bit GGUF 372–475 GB · 2-bit 239 GB.

744B

TOTAL PARAMS, MOE

40B

ACTIVE PER TOKEN

768 GB

VRAM, EIGHT CARDS

FP8THE CHECKPOINT,
UNQUANTIZED**1M**

CONTEXT TOKENS

LOCAL · BOUGHT ONCE**\$181,858**

The FP8 server: eight cards in 4U as specced below; Newegg street prices on October 7, 2026.

CLOUD · HELD WARM**\$34,300**/mo

Northflank nf-gpu-b200-180-8g, 8 × B200 180 GB; list price, 24/7, CPU and RAM bundled.

01 Local — The FP8 server: eight cards in 4U

Eight 96 GB Blackwell cards in a 4U chassis hold the 744 GB FP8 checkpoint in VRAM, the same way the cloud node does and with the same launch line. Six cards (576 GB) cannot hold it; eight is the minimum, and the remaining ~24 GB carries an FP8 KV cache for a handful of long conversations at a time. Every part is sold by Newegg today.

PART	PICK · WHY IT'S HERE	STREET	BUY
CHASSIS	ASUS ESC8000A-E13-32W, 4U 8-GPU server barebone (dual EPYC 9005) Eight double-width GPU bays with the airflow and redundant 3 kW supplies to match; board, backplane and power are included. In stock at Newegg; the E12 generation is not.	\$8,470 in stock	Newegg ↗
CPU × 2	AMD EPYC 9355, 32 cores (5th Gen, SP5) × 2 Two sockets for 24 memory channels and 256 PCIe 5.0 lanes feeding eight x16 cards; the GPUs do the work. The 48- and 64-core parts are out of stock and unnecessary here.	\$6,188 \$3,094.00 each in stock	Newegg ↗
MEMORY	Kingston Server Premier 64 GB DDR5-5600 ECC RDIMM × 12 = 768 GB × 12 768 GB matches the VRAM, so the whole checkpoint page-caches for fast reloads; one DIMM on six of the twelve channels per socket. Memory is where the money went this year.	\$28,200 \$2,349.99 each in stock	Newegg ↗
GPU × 8	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB × 8 8 × 96 GB = 768 GB, FP8 weights in VRAM. The 300 W blower was made for dense boxes; the passive Server Edition is on Newegg only through a marketplace seller at \$30,000, so the Max-Q it is.	\$136,000 \$16,999.99 each in stock	Newegg ↗
STORAGE	Samsung 9100 PRO 4 TB, NVMe Gen5 × 4 × 4 The FP8 checkpoint alone is 744 GB; four drives hold it plus two quant tiers and the BF16 shards for conversion.	\$3,000 \$749.99 each in stock · list \$1,360	Newegg ↗
All-in, as specced · 5 parts, 5 priced live from newegg.com on October 7, 2026		\$181,858	

Newegg street prices as read on October 7, 2026; the live table at flatclaw.org/builds refreshes hourly. Used-market and unlisted parts are marked. Verify at checkout.

02 Cloud — the same checkpoint, one service

NORTHFLANK · RECOMMENDED · nf-gpu-b200-180-8g

\$34,300/mo · \$47/hr

8 × B200 180 GB · 1,536 GB. The fast node: HBM3e at about 8 TB/s per card, roughly twice the H200, so decode per stream is about twice as fast; FP8 weights and an FP8 KV cache with room for the full 1M-token window. SGLang --tp 8 --enable-moe-ep; the weights-server + customEntrypoint pattern from the 31B box with a ≥1.5 TB volume and a 20–40 minute cold boot (744 GB fetched).

NORTHFLANK · nf-gpu-h200-141-8g

\$18,300/mo · \$25/hr · spot \$14.6/hr ≈ \$10,700/mo

8 × H200 141 GB · 1,128 GB. The economy node: the same checkpoint and launch line at about half the decode speed; context up to 256K with KV room.

NORTHFLANK · nf-compute-400-16 (control plane)

\$144/mo

none. Portal plus one OpenClaw gateway per user, the same service as the standard tenant. The model class does not change the control plane.

NORTHFLANK · weights-server (CPU) + ≥1.5 TB volume

\$60/mo

none. nginx autoindex over the checkpoint directory; the GPU node fetches it at boot.

ALL-IN

About \$34,500 a month held warm on the B200 node and about \$18,500 on the H200 node, control plane and weights-server included; spot on H200 about \$10,700. Schedule the flagship for live load and keep a smaller class as the floor.

THE SAME SHAPE ELSEWHERE

- ▶ **Azure** · ND GB200 v6 (8 × B200) for speed and the 1M window · ND H200 v5 (8 × H200) — One 8-GPU deployment on AKS (NVIDIA device plugin, --tp 8) behind an internal ingress, or an Azure ML online endpoint; weights in Blob, fetched on boot; no public egress.
- ▶ **Azure, cheaper** · NC A100 v4 × N — Ampere has no FP8: a quantized GLM-5.2 only, the same caveat as the two-card workstation.

NOTES

- ▶ Cost reality: this is a different tier. The B200 node is about seventeen times the 31B box held warm, so the frontier class is scheduled for live load and scaled to zero between sessions.
- ▶ Per GPU-hour on Northflank: H100 \$2.74 · RTX PRO 6000 96 GB \$3.00 · H200 ~\$3.14 · B200 ~\$5.87.
- ▶ Routing is unchanged: OpenClaw points at one endpoint; a LiteLLM or vLLM router in the same tenancy adds failover to the H200 node, a quantized fallback or a second region.

03 The quant ladder — what fits where

TIER	FOOTPRINT	FITS
BF16 full precision	~1,500 GB	cloud only, multi-node
FP8 (the prod checkpoint)	~744 GB	eight 96 GB cards locally (short context) · the cloud node with KV room
8-bit	810 GB	GPU server or cloud
5-bit	570 GB	512 GB RAM + 192 GB VRAM, tight
4-bit (UD-Q4_K_XL), the workstation	372–475 GB	the 512 GB two-card box, near-lossless
3-bit	290–360 GB	a 384 GB box
2-bit (UD-IQ2_M)	239 GB	a 256 GB box; ~82% top-1 at 84% smaller
1-bit	223 GB	tight budgets; the biggest quality hit

04 Run it

Launch

```
python -m sglang.launch_server --model-path zai-org/GLM-5.2 --tp 8 --quantization fp8 --enable-moe-ep --kv-cache-dtype fp8_e5m2 --c
ontext-length 131072 --port 8000
```

- ▶ 8 × 300 W plus two 280 W CPUs and the platform: about 3.2 kW, a dedicated 240 V circuit and a room that can shed the heat.
- ▶ Same shape as the cloud node, which is the point: the launch line is identical, only the owner changes. Per-stream speed follows memory bandwidth, GDDR7 at 1.8 TB/s per card against the B200's 8 TB/s, so the cloud node decodes faster; this box is FP8 parity you own.
- ▶ Six cards (576 GB) cannot hold the FP8 checkpoint at all, and eight leave about 24 GB for KV. For FP8 at full concurrency and the 1M window, the cloud node's 1,536 GB is the shape.

05 The two-card 4-bit workstation (the cheaper path)

Two Blackwell cards hold attention, the shared layers and the KV cache at CUDA speed; 512 GB of 8-channel DDR5 holds the cold experts of a 4-bit copy. Decode lands around 10–18 tokens per second, prefill is CUDA-fast. Every part is on Newegg today.

PART	PICK · WHY IT'S HERE	STREET	BUY
GPU × 2	NVIDIA RTX PRO 6000 Blackwell Max-Q, 96 GB × 2 Attention, dense layers, the FP8 KV cache and as many hot expert tensors as fit, split across both cards. Two Max-Q blowers at 300 W each sit side by side and run off a 1,600 W supply; the 600 W Workstation Edition pair would want 2,000 W and more airflow.	\$34,000 \$16,999.99 each in stock	Newegg ↗
CPU	AMD Ryzen Threadripper PRO 7975WX, 32 cores Eight DDR5 channels (~325 GB/s) feed the experts, which is the real bottleneck, plus 128 PCIe 5.0 lanes for two x16 cards and the offload engine.	\$3,800 in stock	Newegg ↗
MAINBOARD	ASUS Pro WS WRX90E-SAGE SE sTR5 / WRX90: seven PCIe 5.0 x16 slots, eight DDR5 RDIMM slots, 10 GbE. The only desktop platform with both the lanes and the memory channels.	\$1,205 in stock · list \$1,300	Newegg ↗
MEMORY	Kingston Server Premier 64 GB DDR5-5600 ECC RDIMM × 8 = 512 GB × 8 Holds the whole 4-bit GGUF (~475 GB) in RAM; experts stream to the GPUs per token. All eight channels populated: bandwidth sets decode speed. Memory prices roughly quadrupled since the June 2026 sheet; this line is where the money went.	\$18,800 \$2,349.99 each in stock	Newegg ↗
POWER	Seasonic PRIME TX-1600, 1600 W, 80+ Titanium 2 × 300 W Max-Q + 350 W CPU + platform ≈ 1.25 kW continuous. Native 12V-2x6 cables for both cards.	\$733 in stock	Newegg ↗
CPU COOLER	SilverStone XE360-TR5-V2, 360 mm AIO Full sTR5 cold-plate coverage for the 350 W Threadripper while two GPUs dump heat into the case.	\$386 in stock · list \$430	Newegg ↗
CASE	Fractal Design Meshify 2 XL Mesh airflow and room for two long cards and an E-ATX / SSI-EEB board. Alternative: a 4U rack chassis if it lives in a closet.	\$240 in stock	Newegg ↗
STORAGE	Samsung 9100 PRO 4 TB, NVMe Gen5 × 2 × 2 GGUF tiers run 223–810 GB each; Gen5 keeps mmap and load fast. Two 4 TB drives hold several quant levels side by side (the 8 TB model is not listed).	\$1,500 \$749.99 each in stock · list \$1,360	Newegg ↗
All-in, as specced · 8 parts, 8 priced live from newegg.com on October 7, 2026		\$60,664	

Launch

```
./llama-server -m GLM-5.2-UD-Q4_K_XL.gguf -ngl 99 -n-cpu-moe 999 -ts 50,50 -c 131072 --flash-attn --cache-type-k q8_0 --cache-type-v
q8_0 --host 0.0.0.0 --port 8000
```

Alternative

```
ktransformers --model zai-org/GLM-5.2 --gguf_path ./GLM-5.2-UD-Q4 --cpu_infer 32 --max_new_tokens 131072 --port 8000
```

- ▶ The engine keeps attention and KV on the cards and routes cold experts to RAM: llama.cpp (-ngl 99 -n-cpu-moe 999 -ts 50,50 --flash-attn --cache-type-k q8_0 --cache-type-v q8_0) or KTransformers for more expert throughput. OpenClaw points at the single OpenAI endpoint, exactly like the 31B box.
- ▶ A 744B MoE owns all the memory: no room for image or voice on this box; run modality on a second machine as the standard build does.
- ▶ This is the cheaper path: a quantized GLM-5.2 (4-bit is excellent) at workstation speed. FP8 parity you own is the eight-card server above.